

BANDA JAIRAJ

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OBJECTIVE

Final-year ECE student focused on embedded firmware. Designed and built a quadcopter flight controller end to end (schematic, PCB, firmware) and wrote bare-metal STM32 drivers at register level. Comfortable debugging C/C++ and Python from hardware to application layer. Seeking an entry-level embedded/firmware role.

EDUCATION

B.Tech, ECE — Keshav Memorial College of Engineering (JNTUH), Hyderabad | CGPA: 6.05 | Expected 2027

Intermediate, MPC — Narayana Junior College, Hyderabad | 85% | Passed out 2023

Schooling, ICSE — Sri Sai Public School, Habsiguda, Hyderabad | 80% | Passed out 2021

TECHNICAL SKILLS

Languages: Embedded C, C++, Python

Embedded Platforms: ESP32, ESP8266, STM32 (BlackPill F411CEU6), Arduino, sensor interfacing, PCB design and bringup

IoT & Software: MQTT/HTTP, Blynk, ThingSpeak, Node.js/Express, SQLite, Telegram Bot API, Google Gemini API

Other: ROS 2 (in progress)

PROJECTS

Quadcopter Flight Controller | *Personal Project* | *Embedded Firmware, PCB*

- Sole developer across schematic, PCB, and firmware bring-up for a quadcopter flight controller; tested firmware behaviour on the physical hardware.

STM32 Bare-Metal GPIO & Button Input Drivers | *Personal Project* | *STM32F411, Embedded C*

- Wrote LED-blink and button-input drivers for the STM32F411 (BlackPill) using direct register manipulation only, with no HAL or CMSIS.
- Verified peripheral behaviour against the reference manual register map.

ESP32-Based RC Car (Motor Control System) | *Personal Project* | *ESP32, C++*

- Implemented PWM-based DC motor control using TB6612FNG/DRV8833 drivers with Blynk remote control; debugged driver and timing issues through incremental functional testing.

IoT Weather Station & Cloud Dashboard | *Industrial Training Project* | *ESP32, DHT22, ThingSpeak*

- Built DHT22 sensing with Wi-Fi upload to a ThingSpeak dashboard and threshold-based email alerts; analysed the logged data in MATLAB.

Ultrasonic Radar System | *College Project* | *ESP32, C++*

- Built a servo-driven ultrasonic radar sweep, debugging sensor-noise and timing issues to produce a reliable distance-vs-angle scan.

ESP32 Home Automation System | *College Project* | *ESP32, Blynk, IoT*

- Developed remote monitoring and control through the Blynk app; tested connectivity and command reliability over Wi-Fi.

BUNNY — Personal Assistant Server | *Personal Project* | *Python, Termux, SQLite, Telegram Bot API*

- Converted an old Android phone into a lightweight always-on server using Termux, Python, and SQLite, designed as a low-footprint coordinator that calls cloud APIs.
- Built a job-search pipeline with scheduled searches, scoring and filtering of listings, and delivery through a Telegram bot.

NARADH — Personal AI Task Router | *Full-Stack Web App* | *Node.js/Express, SQLite, Google Gemini API*

- Built a full-stack app that analyses a prompt and routes it to the most suitable of 8 AI platforms, using the Gemini API for intent classification.

CERTIFICATIONS

- JIJNASA National Certificate
- Aigen Labs Certification
- Outskill Certification
- KMCE Certification